

UTM TENSILE REPORT — UTM_Test_20260817_155639

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A_0) · anchor self-calibration

19.9 MPa

UTS

19.5 MPa

Yield σ_y

1.22 GPa

Modulus E

4.0 %

Failure ϵ_f

676 kJ/m³

Toughness

305 N

Preload anchor

TEST SETTINGS

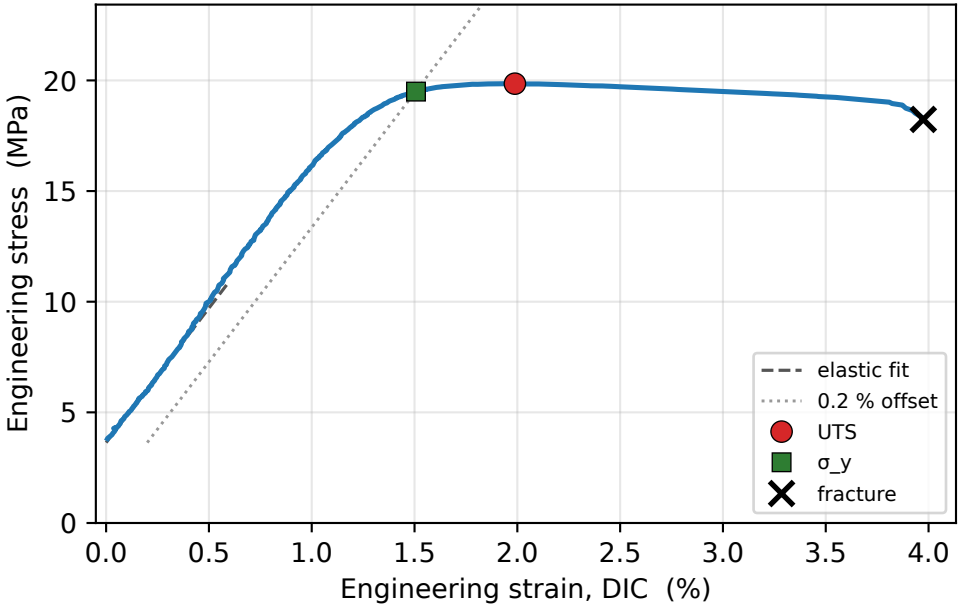
Specimen	UTM_Test_20260817_155639
DIC mode	White
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 329.264
Comment	—
Test date	2026-08-17 15:54:02
Duration	162.3 s

VALIDATION vs REFERENCES

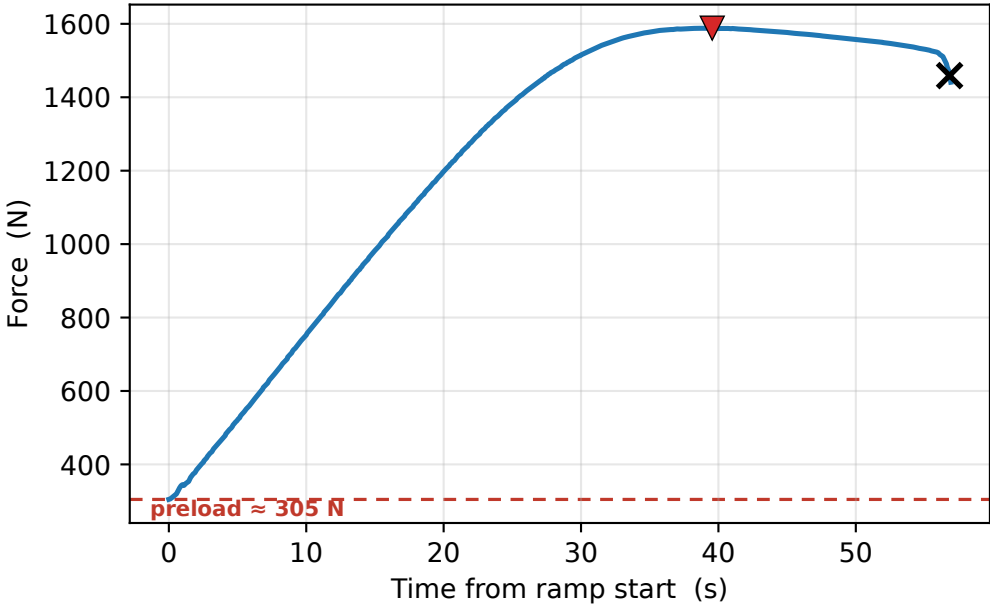
prop	meas	Chacón	E-PLA k
UTS	19.9	32-60 low	×2.92
σ_y	19.5	30-50 low	×2.98
E	1.2	3-5.5 low	×2.36
ϵ_f	4.0 %	(range 3-7 %)	×2.01

⚠ strength outside journal range

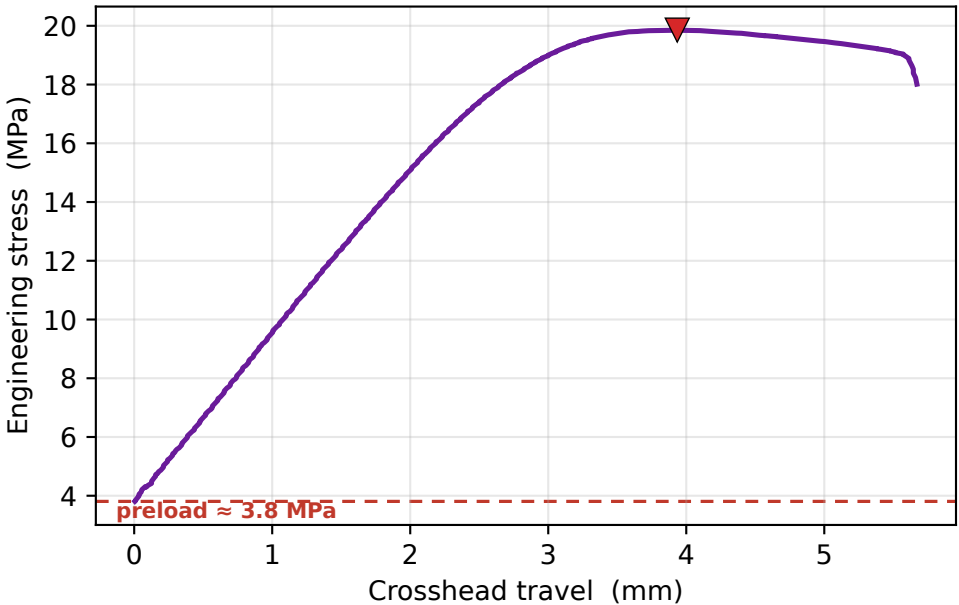
Stress-strain



Load vs time



Stress vs crosshead



DIC strain vs crosshead — the compliance gap

